LOMBA KOMPETENSI SISWA
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TEST PROJECT
MODUL A – CIENT SERVERS

BIDANG LOMBA

TEKNOLOGI INFORMASI SISTEM ADMINISTRASI JARINGAN

IT NETWORK SYSTEMS ADMINISTRATION

DESCRIPTION OF PROJECT

A small startup company's production environment contains numerous services within multiple Operating Systems. We will ask you to install, configure, and integrate the different services. You will have two work days to complete two different sets of tasks. Each set of tasks is to be completed within one work day. For the second work day, you will continue the work of another engineer, who should have fully configured systems according to the day one set of tasks. You may check, change, or update the existing systems, but we will only score the tasks you're supposed to do on the second work day.

EDGE ZONE

NAT and Routing

- Enable port NAT to all private addresses in majapahit.id, and pasai.org so that every device can connect to public subnets.

DHCP

- Access FW and install any dhcp server. Configure to serve DHCP in majapahit.id.
  - Network: 172.16.0.0/20
  - Available Addresses: 172.16.0.1 - 172.16.14.253
  - Gateway: 172.16.15.254
- Create static lease for srv1 and srv2. Configure those servers to use DHCP.

Load Balancer

- Access FW and install haproxy. Configure it to loadbalance web requests.
  - Use /etc/haproxy/haproxy.cfg configuration file.
  - Do not change default configuration values.
- Load Balance www.majapahit.id with round robin algorithm. Name the backend sabang_www.
- Load Balance 100 user websites with source ip algorithm. Name the backend sabang_user
  - user001.public.majapahit.id
  - user002.public.majapahit.id
  - user003.public.majapahit.id
  - …
  - user099.public.majapahit.id
  - user100.public.majapahit.id

Email Services

- Install any mail service and configure a sending-only mail server in fw.majapahit.id
  - Secure SMTP Port 465
  - Enable PAM Authentication
    - Make sure all local users are able to authenticate
  - Use the self-signed certificate from Windows CA for this task.

INTERNAL ZONE

Basic Configuration

- Configure IP Address of all Linux devices according to the addressing table.
    - Create SSH user 'patah' with password specified in the appendix.
    - Enable SSH access from any devices.

Sabang DNS

- Access srv1 and srv2 and install bind9. Configure it to serve domain majapahit.id and indonesia.com using the private addresses.
- Create A records of srv1.majapahit.id, srv2.majapahit.id, and fw.majapahit.id that points to their respective addresses.
- Create NS records:
    - ns1.majapahit.id that points to address of srv1.majapahit.id
    - ns2.majapahit.id that points to address of srv2.majapahit.id
- On both name servers, create the following subdomains using address record:
    - www.majapahit.id that points to all addresses of fw.majapahit.id
    - 100 user subdomain that points to all addresses of fw.majapahit.id.:
        - user001.public.majapahit.id
        - user002.public.majapahit.id
        - user003.public.majapahit.id
        - …
        - user099.public.majapahit.id
        - user100.public.majapahit.id

Sabang Web Hosting

- Access srv1 and srv2 and install apache2. Configure it to serve all websites in majapahit.id.
  - Install curl to test the web service.
- Create /var/www/index.html default page with content specified in Appendix.
- Create /var/www/userXXX/index.html default page for 100 users with content specified in Appendix. Example:
  - /var/www/user001/index.html
  - /var/www/user002/index.html
  - /var/www/user003/index.html
  - …
  - /var/www/user099/index.html
  - /var/www/user100/index.html
- Create virtual host www.majapahit.id serving /var/www/index.html
- Create 100 virtual hosts userXXX.public.majapahit.id serving /var/www/userXXX/index.html . Example:
  - user001.public.majapahit.id serving /var/www/user001/index.html
  - user002.public.majapahit.id serving /var/www/user002/index.html
  - user003.public.majapahit.id serving /var/www/user003/index.html
  - …
  - user099.public.majapahit.id serving /var/www/user099/index.html
  - user100.public.majapahit.id serving /var/www/user100/index.html

HTTPS Access

- Use DC from Windows to generate self-signed certificates required for this task.
- Enable HTTPS in the load balancer serving www.majapahit.id.
- Enable HTTPS in the web server serving user websites.
  - Configure load balancer to forward traffic non-terminated HTTPS. Load balancer will not handle HTTPS, HTTPS is handled by the backend web server.
  - Only enable to the first 5 and last 5 users :
    - user001.public.majapahit.id
    - user002.public.majapahit.id
    - user003.public.majapahit.id
    - user004.public.majapahit.id
    - user005.public.majapahit.id
    - user096.public.majapahit.id
    - user097.public.majapahit.id
    - user098.public.majapahit.id
    - user099.public.majapahit.id
    - user100.public.majapahit.id

Sabang Failover

- Access srv1 and srv2 and install keepalived. Configure it to serve as a failover when one of the servers is down.
    - Set srv2 as backup.
    - Use virtual IP 172.16.1.205
    - Use VRRP ID 205
    - Use script to check DNS service. When DNS service is down, move virtual IP to another server.

Sabang iSCSI

- Access srv1 and srv2 and install tgt. Configure it to serve iSCSI target in majapahit.id.
    - Use device /dev/sdb, /dev/sdc, /dev/sdd, /dev/sde, /dev/sdf, /dev/sdg, /dev/sdh, /dev/sdi, /dev/sdj, /dev/sdk
    - A total of 20 disks will become iSCSI targets
- Access fw.majapahit.id and install open-iscsi. Configure it to connect to iSCSI target in srv1 and srv2.
    - Don't create any filesystem on the disk.
    - Make sure both disks from srv1 and srv2 are available on fw.majapahit.id.

Folder Backup

- Backup all users virtual host folder in srv1 and srv2 to C:\public in FILES.
    - Backup all 100 users folder.
        - /var/www/user001/
        - /var/www/user002/
        - /var/www/user003/
        - …
        - /var/www/user099/
        - /var/www/user100/
    - Mount the remote backup folder locally at /backup.
        - For server srv1, place backup content in directory /backup/srv1/
        - For server srv2, place backup content in directory /backup/srv2/
    - Schedule backup every 1 hour using cron as root user. It is okay to overwrite previous backups.

DMZ ZONE

Basic Configuration

- Configure IP Address of all windows servers according to the addressing table.
- Determine and set gateway IP address to all windows servers.
- Configure hostname of all windows servers.
- Make all windows servers pingable from any devices.

File Sharing

- Create and Share Folder C:\backup\ in host FILES
  - Enable AD Authentication, permit user 'Administrator' only
- Create and Share Folder C:\public\ in host FILES
  - Disable Authentication, anonymous user can read and write to this folder.

Cross Platform iSCSI

- Configure iSCSI Initiator on FILES
- Connect to all 20 disks in Sabang iSCSI.
- Create an NTFS filesystem and mount all 20 disks to E:/, F:/, and so on in any order.
  - Make sure 20 extra disks is accessible via File Explorer

Web Service HTTPS

- Access FILES and Install IIS web service.
- Serve web https://www.pasai.org
  - Serve file index.html as specified in the appendix
  - Use a Self-Signed Certificate from LKSN2021-CA to enable HTTPS.
- Serve 25 user websites:
  - Make sure it is accessible using following URL https://user01.pasai.org, https://user02.pasai.org, https://user03.pasai.org, and so on until https://user25.pasai.org.
  - Use a Self-Signed Certificate from LKSN2021-CA to enable HTTPS.
  - Serve file users.html in the following directory:
    - C:\www\users\01\users.html
    - C:\www\users\02\users.html
    - C:\www\users\03\users.html
    - …
    - C:\www\users\25\users.html

Active Directory and DNS

- Access DC and Configure AD for domain pasai.org
- Create a normal AD user 'patah' with password specified in the appendix.
- Create DNS Address Record for servers in pasai.org according to their hostnames, for example DC.pasai.org
- Create DNS Record www.pasai.org pointing to IP Address of STORAGE
- Create 25 DNS Records for users website pointing to IP Address of STORAGE:
  - user01.pasai.org
  - user02.pasai.org
  - …
  - user25.pasai.org

VPN

- Access FILES and Install RRAS and NPS.
- Configure VPN Server for Windows Client
    - Authenticate using username and password in Active Directory.
    - Permit any users in the domain to authenticate.
- Make sure client can connect to the VPN using FW's Public IP.
    - Create the VPN in client with the name 'Public' so that the user can connect immediately.
    - Access the FW and open the ports needed.

Certificate Authority

- Access DC and Configure DC to issue required Certificates by Linux Services
    - Common Name : LKSJATENG-CA
    - Do not join this server to any domain.
- Generate certificates required by other services with the following DNS name:
    - www.majapahit.id
    - *.public.majapahit.id
    - www.pasai.org
    - *.pasai.org
    - Save these certificates and their private keys as one pfx file in C:\cert\
    - Use **Skills39** as export password
    - Use the DNS name as filename :
        - www.majapahit.id.pfx
        - public.majapahit.id.pfx
        - www.pasai.org.pfx
        - wildcard.pasai.org.pfx

Backup Configuration

- Access DC and Schedule Backup using Windows Backup.
- Backup folder C:\cert and all its contents to \\FILES\backup right away.
    - Create at least one empty .txt file to the backup folder
    - Schedule the backup every day at 1 AM.

APPENDIX

Users and Passwords

Username	Password	Remark
patah	KesultananDemak2021	New User
user	Skills39	Debian only
Administrator	Skills39	Windows Server only
root	Skills39	Debian only
competitor	Skills39	Windows 10 only

End Devices Addressing Table

Device Name	IP Address CIDR	Network
srv1	172.16.1.201/20	majapahit.id
srv2	172.16.1.202/20	majapahit.id
fw	172.16.15.254/20	majapahit.id
	172.10.254.254/19	public.net
	172.17.9.1/22	pasai.org
DC	172.17.10.100/22	pasai.org
FILES	172.17.11.102/22	pasai.org
client	172.10.245.245/19	public.net

/var/www/index.html

```
<h1> Default Page </h1><br>
<br>
This page has not been modified by the owners.
```

/var/www/userXXX/index.html [Replace XXX in the folder name and file content with user number, for example /var/www/user009/index.html]

```
<h1> Default Page for userXXX </h1><br>
<br>
This page has not been modified by the user.
```

TOPOLOGY

